

PHRONESIS: Complete Application Plan Document

Version: 1.0

Date: August 31, 2026

Status: Design Phase - Ready for Development Planning

Platform: React + TypeScript + Vite + Tailwind + Supabase

Target Markets: Uganda / Africa-wide

1. EXECUTIVE SUMMARY

Phronesis is an AI-powered vehicle diagnostic and advisory platform with two distinct user journeys:

1. **Pre-Car Ownership Journey** — For users researching, comparing, and shopping for vehicles
2. **Post-Car Ownership Journey** — For vehicle owners managing diagnostics, maintenance, and repairs

The app features an adaptive AI avatar (Phronesis) that serves as an intelligent conversational router, guide, and diagnostician. Users never manually navigate; instead, the avatar interprets intent from natural language (text, voice, video) and takes them where they need to go.

Core Innovation: Context-aware, intent-driven navigation powered by conversational AI + real-time OBD data + community mechanic marketplace.

2. USER HANDLING & AUTHENTICATION

2.1 Authentication Methods

- **Sign Up** (email + password)
- **Sign In** (email + password)
- **Google Sign In** (OAuth)

2.2 Session Persistence

- Sessions stored in Supabase Auth
- Tokens cached securely on device
- Auto-refresh on app launch

2.3 User Profile Data

- Email, username, phone number
- Profile picture
- Primary journey (Pre-car or Post-car)
- Secondary journey access (toggle-able)
- Preferences (language, accent, notification settings)

2.4 Chat History & Continuity

- All conversations stored (Supabase Chat History table)
- One continuous thread per user (not separate conversations)
- Avatar remembers full context across sessions
- Searchable/filterable history (future phase)

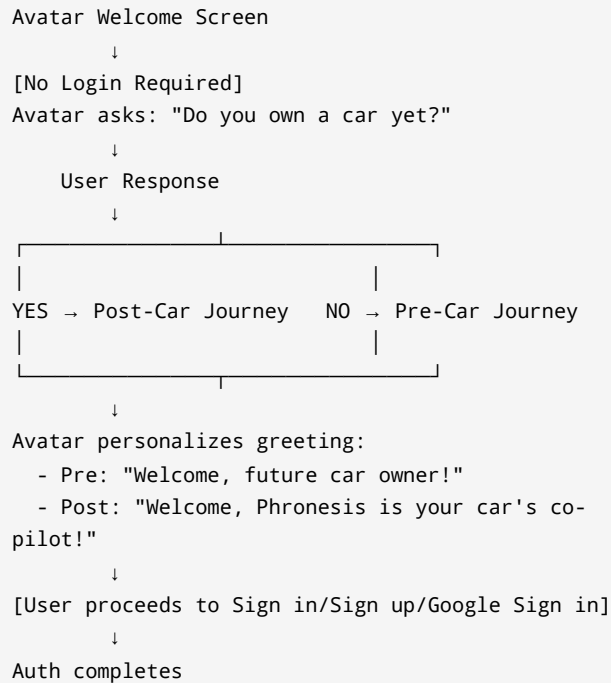
2.5 Account Management Page

- Edit profile picture

- Edit username (Avatar uses this in conversations)
 - Edit/add phone number
 - Edit/add email
 - Toggle between Pre-car and Post-car journeys
 - Manage connected OBD devices
 - View chat history
 - Privacy & data settings
-

3. JOURNEY DETECTION & ROUTING

3.1 Onboarding Flow



↓

Avatar continues conversation → Home/Chat page
(for their primary journey)

3.2 Journey Access

- Users have one **primary journey** (inferred from onboarding)
- Can access **both journeys** via toggle in Account page or Nav
- Avatar adapts personality based on active journey

3.3 Dynamic Journey Refinement

- User can ask cross-journey questions
- Avatar validates context and asks clarifying questions
- Example: "I heard a weird noise" (Post-car question) while in Pre-car journey → Avatar asks "Do you own this car?" → Routes appropriately

4. AVATAR DESIGN & BEHAVIOR

4.1 Avatar Identity

- **Name:** Phronesis
- **Form:** Malleable, fluid orb (not static shape)
- **Personality:** Friendly, helpful, approachable
- **Gender:** Male
- **Accent:** African (Uganda-based; localized by country)
- **Consistency:** Same character across both journeys, adapts personality mode

4.2 Personality Modes

Mode	Journey	Tone	Behavior
Research Mode	Pre-Car	Advisor, curious, exploratory	Helps explore options, compare, learn
Diagnostic Mode	Post-Car	Analytical, problem-solver, reassuring	Diagnoses issues, prioritizes fixes, recommends solutions

4.3 Avatar Appearance & Expression

- **Visual Changes:**
 - Expressions/emotions shift during conversations
 - (Post-Car only) Reflects car health status
 - Glows green/healthy when car is in good condition
 - Dims or shows concern when issues detected
 - Animates with urgency based on problem severity
- **Fluid Orb Animation:** Continuous subtle morphing/pulsing

4.4 Avatar Positioning (Responsive Layout)

Screen	Position	Status
Avatar Welcome	Center	Full focus
Home/Chat	Center	Full focus

Screen	Position	Status
Sign in/Sign up	Corner (minimized)	Still interactive
Diagnosis Page	Corner (minimized)	Still interactive
Solutions Page	Corner (minimized)	Still interactive
Maps & Navigation	Corner (minimized)	Still interactive
Account Page	Corner (minimized)	Still interactive

Core Principle: Avatar adapts to screen content – centers when it's the focus, minimizes when other information needs space. User can still tap/interact with corner avatar.

4.5 Avatar Voice & Localization

- **TTS Provider:** Google Cloud Text-to-Speech or Azure Speech Services (supports African accents)
- **Accent:** Localized per country (Uganda, Kenya, Nigeria, South Africa, etc.)
- **Tone:** Adapts based on context (encouraging, serious, playful, etc.)

5. PRE-CAR OWNERSHIP JOURNEY

5.1 Purpose

Guide users through vehicle research, comparison, pricing, and purchase decisions before buying.

5.2 Services

5.2.1 Car Information Library

- **Data included:** Engine type, parts catalog, country of manufacture, mileage, fuel consumption specs, tank size, transmission type, dimensions, safety features, warranty info
- **Accessibility:** Browse by make/model/year or ask Avatar ("Tell me about Honda Civic engines")
- **Interaction:** Text, voice, or upload images (user can point camera at a car)

5.2.2 Car Recommendations

- Avatar asks budget constraints, preferences, use case
- Recommends vehicles matching criteria
- Compares up to 3 vehicles side-by-side
- Shows pros/cons for each

5.2.3 Price Gauging & Market Analysis

- Shows current market price ranges (by region, condition, year)
- Alerts on deal opportunities
- Trends over time (prices rising/falling)
- Regional pricing differences (e.g., "Same car is cheaper in Nairobi than Kampala")

5.2.4 Multi-Modal Assistant Interaction

- **Text input:** Type questions to Avatar

- **Voice input:** Speak to Avatar (STT)
- **Video recording:** Show Avatar a car's condition, damage, or interior (Avatar analyzes)

5.2.5 Service Location Mapping

- Shows dealerships, private sellers, auto shops near user
- Real-time navigation to locations
- Reviews/ratings of dealerships
- Can filter by vehicle type, price range, distance

5.3 App Flow (Pre-Car Ownership)

```
Avatar Welcome
↓
Sign in/Sign up/Google Sign in
↓
Home/Avatar Chat
(User asks: "What car should I buy under $5000?")
↓
Avatar interprets → Routes to Car Info &
Recommendations
↓
Car Info & Recommendations Page
(Shows matching vehicles, specs, pricing)
↓
User asks: "Where can I find these cars?"
↓
Avatar routes to Maps & Navigation
↓
Maps & Navigation Page
(Shows nearby dealerships/sellers)
↓
User asks: "Tell me more about my profile settings"
```


↓
Avatar routes to Account Page

6. POST-CAR OWNERSHIP JOURNEY

6.1 Purpose

Help vehicle owners diagnose problems, find solutions, manage maintenance, and access repair services.

6.2 Services

6.2.1 Real-Time Avatar Assistant

- Always available for questions about car health, maintenance, repair costs, service providers
- Remembers user's car(s) and history

6.2.2 OBD Bluetooth Monitoring

- Connects to OBD-II adapter via Bluetooth
- **Data collected:**
 - Engine fault codes (DTCs)
 - Real-time sensor readings (temperature, RPM, fuel consumption, O2 levels, etc.)
 - Battery voltage
 - Vehicle speed
 - Emission data
- **Real-time polling:** Continuous monitoring during drive or on-demand

- **Alerts:** Avatar notifies user of anomalies ("Engine temp is rising," "Check engine light triggered")

6.2.3 Real-Time Notifications & Alerts

- Critical issues (engine failure risk, brake problems)
- Warning issues (unusual readings, maintenance due)
- Informational (fuel efficiency tips, service reminders)
- User can customize alert thresholds

6.2.4 Multi-Modal Assistant Interaction

- **Text input:** Type symptoms ("Rough idle at cold start")
- **Voice input:** Describe issue verbally (STT)
- **Audio recording:** Record engine/car sounds for Avatar analysis
- **Video recording:** Film visible damage, dashboard, warning lights
- **OBD data:** Real-time diagnostic data from adapter

6.2.5 3D Car Hologram Visualization

- **Display:** Interactive 3D model of user's car (make/model/year specific)
- **Features:**
 - Zoom in/out on specific areas
 - Select individual parts to inspect
 - Highlight damaged/problematic areas (based on diagnosis)
 - Color-coded urgency (red=critical, yellow=warning, green=healthy)

- Part names and specs on hover
- **Integration:** Updates in real-time as Avatar diagnoses issues
- **Use case:** User can visualize exactly where a problem exists ("That's the alternator")

6.2.6 Diagnosis & Repair Solutions

- Avatar analyzes OBD data + user input
- Generates **diagnosis report:**
 - Identified issues (with confidence levels)
 - Root cause analysis
 - Urgency ranking (Critical/High/Medium/Low)
 - "Nice-to-have" vs. "Must-fix" prioritization
- Shows **repair cost estimates** and recommends solutions

6.2.7 Mechanic Service Finding & Marketplace

- **Mechanic Database:** Custom-built partner shop directory
- **Ranking:** Shops ranked by user reviews (best to least wanted)
- **Smart Matching:** Avatar suggests best nearby mechanics based on:
 - Specific car issue diagnosed
 - Shop ratings/reviews
 - Proximity to user
 - Specialization (some shops specialize in certain brands)
- **Cost Options:**

- Industry cost range (from knowledge base)
- Actual shop quotes (if integrated)
- User contacts shops directly for custom quotes
- **Comparison:** User can compare costs, reviews, and services across multiple providers

6.2.8 Service Reports & Transparency

- **Phronesis Report:** AI-generated diagnosis report (user reference)
- **Mechanic Report:** Generated by service provider after work (future phase)
- **Dual-sided future:** Separate apps for users and service providers (both generate/view reports)
- **Content:**
 - Parts cost vs. labor cost breakdown
 - Urgency level
 - Priority (must-fix vs. nice-to-have)
 - Timeline for repair
 - Warranty/guarantee info

6.2.9 Follow-up & Feedback Loop

- **Issue Resolution:** User marks issue as "resolved" (OBD can verify via retesting)
- **Service Rating:** User rates mechanic service (1-5 stars + comments)
- **Review Distribution:** Rating sent to service provider (affects their reputation)
- **Photo Upload:** User can upload photos of completed work (builds transparency)

- **Learning:** Phronesis learns from every outcome to improve future recommendations

6.2.10 Real-Time Navigation to Services

- **Route Planning:** Directions to recommended mechanic shops
- **Live Navigation:** Google Maps integration with turn-by-turn directions
- **Traffic Awareness:** Avoids congestion (real-time traffic data)
- **Estimated Time:** ETA to service provider

6.3 App Flow (Post-Car Ownership)

```
Avatar Welcome
↓
Sign in/Sign up/Google Sign in
↓
[OBD Pairing Prompt during onboarding - can defer]
↓
Home/Avatar Chat
(User says: "My car's making a weird noise")
↓
Avatar asks: "Can you describe it?" + pulls OBD
data
↓
Avatar routes to Diagnosis Page
↓
Diagnosis Page
(3D hologram shows engine block, audio analysis,
OBD codes)
Avatar displays:
- Identified problem (e.g., "Engine knock
detected")
- Urgency (High)
```

```
- Repair solutions
  ↓
User asks: "How much will this cost to fix?"
  ↓
Avatar routes to Solutions Page
  ↓
Solutions Page
(Shows recommended mechanics, cost range, shop
quotes)
User selects mechanic
  ↓
User asks: "How do I get there?"
  ↓
Avatar routes to Maps & Navigation
  ↓
Maps Page
(Shows route to selected mechanic, real-time
directions)
  ↓
[After service] User marks issue as resolved
  ↓
Avatar prompts feedback (rate mechanic)
  ↓
Avatar learns & updates recommendation system
```

7. CHAT HUB & INTELLIGENT ROUTING

7.1 Home/Avatar Chat Purpose

Universal Q&A space — users ask anything (car-related and platform-related) within their stage context.

7.2 Conversation Continuity

- **Memory:** Avatar remembers everything from previous chats

- **Context:** Full conversation history carried forward across sessions
- **Thread:** One continuous thread per user (not segmented conversations)
- **Learning:** Avatar improves responses based on user history and outcomes

7.3 Context-Aware Validation

Avatar detects intent and asks clarifying questions:

User Input	Avatar Response	Action
"What's that weird noise?" (Pre-car user)	"Are you asking about a car you own or one you're considering buying?"	Route based on answer
"Show me cars under \$8k" (Post-car user)	"Looking to buy another vehicle?"	Confirm intent → Route to Car Info
"Can I get an inspection report?" (Pre-car)	"Let me help you find inspection services"	Route to Maps/Services

7.4 Intent-Driven Navigation

User **never manually navigates**. Avatar routes based on natural language:

Intent	Destination	Trigger
Browse/compare cars, get recommendations	Car Info & Recommendations	"Show me..." / "Recommend..." / "What car..."
Describe car problem/symptom	Diagnosis Page	"Weird noise..." / "Check engine light..." / "Vibration..."
Find mechanics, check costs	Solutions Page	"How much to fix?" / "Find a mechanic..." / "Get a quote..."
Navigate to location	Maps & Navigation	"Where can I find...?" / "Show me directions..."
Manage account	Account Page	"Change my profile..." / "Edit username..."

8. OBD BLUETOOTH INTEGRATION

8.1 OBD Connection Flow

- **Timing:** Pairing requested during onboarding (Sign Up)
- **Optional:** User can complete setup later if they skip initial prompt

- **Reconnection:** If connection drops, user prompted to reconnect

8.2 Bluetooth Permissions

- **Request timing:** Per device OS requirements
 - iOS: Permissions requested on first OBD access
 - Android: Permissions requested during onboarding or on first access
- **If denied:** User can still interact via alternative methods (see 8.3)

8.3 Fallback Modes (No OBD or Permission Denied)

User can still diagnose by providing:

- **Text:** Type symptoms in natural language
- **Voice:** Describe issue verbally (STT)
- **Audio recording:** Record engine/car sounds for analysis
- **Video recording:** Film dashboard, warning lights, damage
- **Manual inspection:** User describes what they see/hear

Result: Diagnosis with lower confidence/accuracy but still functional.

8.4 OBD Data Collection

Data Type	Examples	Usage
Engine Fault Codes (DTCs)	P0101 (Mass Airflow), P0420 (Catalyst)	Identify specific problems

Data Type	Examples	Usage
Sensor Readings	Temperature, RPM, O2 levels, fuel consumption	Real-time monitoring
Vehicle Status	Battery voltage, speed, transmission status	Comprehensive health check
Emissions Data	Particulate levels, NOx	Environmental compliance

8.5 Real-Time Monitoring

- **Polling:** Continuous during vehicle operation or on-demand
- **Avatar Alerts:** "Your engine temp is rising to 110°C" / "Check engine light just triggered"
- **Dashboard:** User can view live OBD metrics (optional)
- **Logging:** All data stored in Supabase for historical analysis

8.6 Graceful Degradation

Scenario	Behavior
OBD disconnects mid-diagnosis	Process halts, user prompted to reconnect; can re-diagnose with new data
No OBD adapter	Manual symptoms + multi-modal input (text/voice/video) → diagnosis with lower confidence

Scenario	Behavior
No Bluetooth permission	Use text/voice/video input only; OBD features unavailable

9. SOLUTIONS PAGE & MARKETPLACE

9.1 Mechanic Partner Database

- **Source:** Custom-built partner directory (grows with user base)
- **Data per shop:** Name, location, phone, hours, specialties, ratings, reviews
- **Ranking:** Shops ranked by user reviews and community feedback (best to least wanted)

9.2 Smart Mechanic Recommendation

Avatar suggests best mechanics based on:

- **Diagnosed issue:** "Your car needs alternator replacement
→ Here are shops specializing in electrical"
- **Location:** Proximity to user
- **Ratings:** User reviews and Phronesis community rating
- **Cost:** Shop pricing for similar jobs (if available)

9.3 Cost Breakdown & Transparency

Service report includes:

- **Parts cost vs. labor cost** (itemized)
- **Urgency level:** Critical / High / Medium / Low

- **Prioritization:** "Must-fix" vs. "Nice-to-have" repairs
- **Timeline:** How soon repair should be done
- **Multiple quotes:** Compare costs across different shops

9.4 Service Report Structure

PHRONESIS DIAGNOSTIC REPORT

Vehicle: 2015 Honda Civic
Issue: Engine Knock
Detected: OBD code P0325
Confidence: 92%

DIAGNOSIS

Root Cause: Low-quality fuel or carbon buildup
Urgency: HIGH
Timeline: Fix within 2 weeks

REPAIR SOLUTIONS

Option 1: Carbon cleaning (labor only)
Cost: \$120-180 (parts: \$0, labor: \$120-180)

Option 2: Replace knock sensor
Cost: \$250-400 (parts: \$80-120, labor: \$170-280)

RECOMMENDED MECHANICS (Top 3)

1. ABC Auto Repair (4.8★) - \$280 total
2. Quick Fix Motors (4.6★) - \$320 total
3. John's Garage (4.4★) - \$250 total

[Contact] [Compare Shops] [Book Service]

9.5 Post-Service Feedback Loop

- **Issue Resolution:** User confirms problem fixed (OBD can re-diagnose to verify)
 - **Mechanic Rating:** 1-5 star review + comments
 - **Review Distribution:** Rating sent to service provider (affects ranking)
 - **Photo Evidence:** User uploads photos of completed work (transparency)
 - **Phronesis Learning:** Outcome feeds into recommendation algorithm
 - "User rated this mechanic highly for knock sensor replacement → boost ranking for electrical issues"
 - "This shop consistently resolves problems → increase priority in recommendations"
-

10. TECHNICAL ARCHITECTURE

10.1 Frontend Stack

- **Framework:** React 18+ (TypeScript)
- **Build tool:** Vite
- **Styling:** Tailwind CSS
- **State management:** React Context or Zustand
- **Real-time communication:** WebSocket (Socket.io or native WebSocket API)
- **Offline support:** Service Workers (cache chat history, maps)

10.2 Backend Stack

- **Primary Database:** Supabase (PostgreSQL)

- **Authentication:** Supabase Auth (email, Google OAuth)
- **Real-time:** Supabase Realtime (WebSocket) or separate WebSocket server
- **Storage:** Supabase Storage (profile pictures, photos of repairs)
- **Functions:** Supabase Edge Functions or Node.js/Python backend

10.3 AI & Conversational Engine

MVP Phase (Starter Model)

- **Hybrid approach:**
 - Claude API (Anthropic) – primary for reasoning, diagnostics, recommendations
 - GPT-4 / GPT-4 Turbo (OpenAI) – fallback and multi-modal scenarios
- **Custom knowledge base:**
 - Africa-wide car specifications (make/model/year/engine/parts)
 - Common issues and solutions (curated from partner mechanics + community)
 - Repair costs (industry data + shop quotes)
 - Maintenance schedules (OEM data)

Future Phase (Custom Model)

- **Fine-tune custom LLM** on accumulated Phronesis data
 - User diagnostics & outcomes
 - Mechanic feedback
 - Successful repair patterns

- **Deploy on own infrastructure** for faster, cheaper inference

10.4 API Integrations (3rd Party)

Service	Purpose	Calls/Use Case
Google Maps API	Location, routing, navigation	Real-time directions to mechanics
Google Cloud TTS/Azure Speech	Avatar voice	Localized accent support (Ugandan, Kenyan, etc.)
Google Cloud STT/Azure Speech	Speech-to-text	User voice input parsing
OBD.js / BluetoothLE APIs	Read OBD adapter	Real-time vehicle diagnostics
Weather API	Environmental context	Road conditions, seasonal maintenance tips
Google Places API	Business info	Mechanic shop details, reviews, hours

10.5 Database Schema (Supabase)

```
-- Users Table
CREATE TABLE users (
  id UUID PRIMARY KEY,
```

```

    email VARCHAR NOT NULL UNIQUE,
    username VARCHAR NOT NULL,
    phone_number VARCHAR,
    profile_picture_url VARCHAR,
    primary_journey VARCHAR ('pre-car' | 'post-car'),
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);

-- Car Profiles Table
CREATE TABLE car_profiles (
    id UUID PRIMARY KEY,
    user_id UUID (FK: users.id),
    vin VARCHAR UNIQUE,
    make VARCHAR,
    model VARCHAR,
    year INT,
    engine_type VARCHAR,
    fuel_type VARCHAR,
    transmission VARCHAR,
    mileage INT,
    tank_size DECIMAL,
    last_service_date DATE,
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);

-- Chat History Table
CREATE TABLE chat_history (
    id UUID PRIMARY KEY,
    user_id UUID (FK: users.id),
    message VARCHAR,
    sender VARCHAR ('user' | 'assistant'),
    timestamp TIMESTAMP,
    context_journey VARCHAR ('pre-car' | 'post-car'),
    attachments JSONB (images/videos/audio)
);

-- OBD Data Log Table
CREATE TABLE obd_data_logs (

```



```

    id UUID PRIMARY KEY,
    user_id UUID (FK: users.id),
    car_profile_id UUID (FK: car_profiles.id),
    dtc_codes JSONB (P0101, P0420, etc.),
    temperature DECIMAL,
    rpm INT,
    fuel_consumption DECIMAL,
    o2_levels DECIMAL,
    battery_voltage DECIMAL,
    timestamp TIMESTAMP,
    alert_triggered BOOLEAN
);

-- Service Reports Table
CREATE TABLE service_reports (
    id UUID PRIMARY KEY,
    user_id UUID (FK: users.id),
    car_profile_id UUID (FK: car_profiles.id),
    diagnosis_text TEXT,
    urgency_level VARCHAR ('critical' | 'high' |
'medium' | 'low'),
    estimated_cost_low DECIMAL,
    estimated_cost_high DECIMAL,
    recommended_solutions JSONB,
    created_at TIMESTAMP,
    resolved_at TIMESTAMP,
    resolution_details TEXT
);

-- Mechanic Partners Table
CREATE TABLE mechanic_partners (
    id UUID PRIMARY KEY,
    name VARCHAR,
    location POINT (geo coordinates),
    phone VARCHAR,
    hours VARCHAR,
    specialties JSONB (electrical, engine,
transmission, etc.),
    avg_rating DECIMAL (1-5),
    num_reviews INT,

```

```

        created_at TIMESTAMP
    );

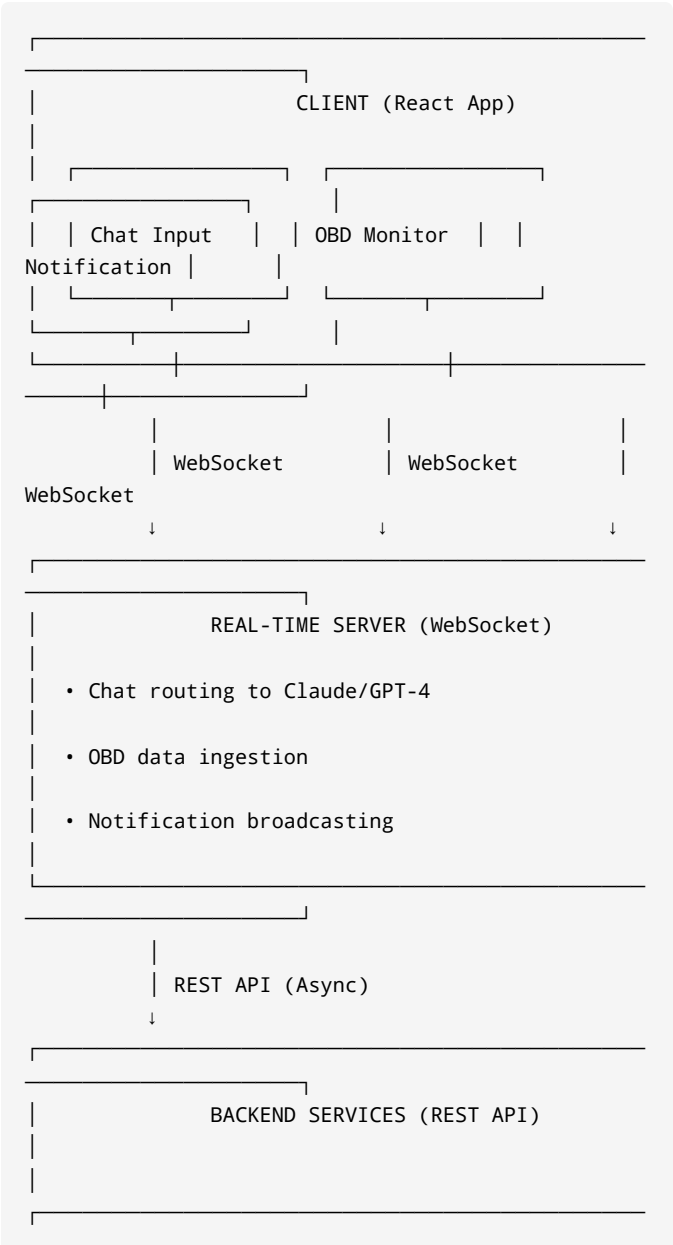
-- Reviews & Ratings Table
CREATE TABLE reviews_ratings (
    id UUID PRIMARY KEY,
    user_id UUID (FK: users.id),
    mechanic_partner_id UUID (FK:
mechanic_partners.id),
    service_report_id UUID (FK: service_reports.id),
    rating INT (1-5),
    comment TEXT,
    photos_url JSONB (array of URLs),
    created_at TIMESTAMP
);

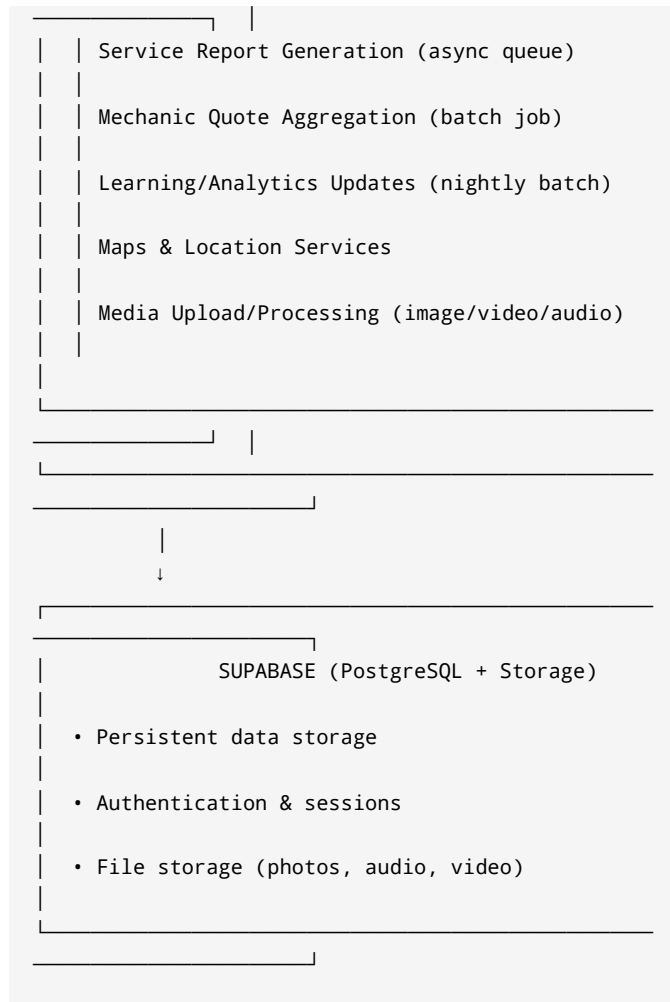
-- Notifications Table
CREATE TABLE notifications (
    id UUID PRIMARY KEY,
    user_id UUID (FK: users.id),
    type VARCHAR ('alert' | 'recommendation' |
'reminder'),
    title VARCHAR,
    message TEXT,
    related_data JSONB,
    read_at TIMESTAMP,
    created_at TIMESTAMP
);

-- OBD Device Connections Table
CREATE TABLE obd_device_connections (
    id UUID PRIMARY KEY,
    user_id UUID (FK: users.id),
    device_name VARCHAR,
    device_id VARCHAR (Bluetooth MAC address),
    paired_at TIMESTAMP,
    last_connected_at TIMESTAMP,
    is_active BOOLEAN
);

```

10.6 Real-Time Architecture





10.7 API Endpoints (REST)

POST	/api/auth/signup	Sign up new user
POST	/api/auth/signin	Sign in user
POST	/api/auth/google	Google OAuth
POST	/api/auth/logout	Logout

POST	/api/car-profiles	Add car profile
GET	/api/car-profiles/:id	Get car profile
PUT	/api/car-profiles/:id	Update car profile
DELETE	/api/car-profiles/:id	Delete car profile
GET	/api/cars/search	Search car info
	(Pre-car)	
GET	/api/cars/recommendations	Get recommendations
	(Pre-car)	
GET	/api/cars/pricing	Get price data
	(Pre-car)	
POST	/api/diagnosis	Run car diagnosis
	(Post-car)	
GET	/api/diagnosis/:id	Get diagnosis report
GET	/api/mechanics/nearby	Find nearby mechanics
GET	/api/mechanics/:id	Get mechanic details
POST	/api/mechanics/:id/quote	Request quote from mechanic
POST	/api/services/:id/rating	Submit service rating
GET	/api/notifications	Get user notifications
PUT	/api/notifications/:id	Mark as read
POST	/api/obd/pair	Pair OBD device
POST	/api/obd/unpair	Unpair OBD device
GET	/api/obd/status	Check OBD connection status
GET	/api/user/profile	Get user profile
PUT	/api/user/profile	Update user

```
profile
DELETE /api/user/account      Delete account
```

10.8 WebSocket Events (Real-Time)

```
// Chat
ws.emit('message:send', { text, attachments })
ws.on('message:receive', { message, timestamp })
ws.on('assistant:typing', {})
ws.on('assistant:stop_typing', {})

// OBD Monitoring
ws.on('obd:connected', { device_name })
ws.on('obd:disconnected', {})
ws.on('obd:data', { dtc_codes, temperature, rpm,
... })
ws.on('obd:alert', { type, message, severity })

// Notifications
ws.on('notification:new', { type, message, data })
ws.on('notification:update', { id, status })

// Typing Indicators
ws.on('user:typing', {})
ws.on('user:stop_typing', {})
```

10.9 Offline Capabilities

- **Chat history:** Cached locally with Service Worker
- **Car info:** Offline car database (MBs, not GBs)
- **Maps:** Downloaded offline maps (Google Maps API w/ offline layers)
- **Sync on reconnect:** All offline actions sync to server when connection restored

10.10 Security Considerations

- **Authentication:** Supabase JWT tokens + refresh tokens
 - **Data encryption:** TLS for all API calls + end-to-end for sensitive chat
 - **OBD data:** Encrypted in transit and at rest
 - **Payment integration (future):** PCI compliance for mechanic service payments
 - **Privacy:** User data not shared with mechanics unless explicitly granted
-

11. DEVELOPMENT ROADMAP

Phase 1: MVP (4-6 weeks)

Goal: Core chat hub + authentication + one journey (Post-car focus)

- [x] User authentication (Sign up, Sign in, Google OAuth)
- [x] Session persistence
- [x] Avatar Welcome screen
- [x] Home/Chat page
 - Text & voice input
 - Basic conversation routing
- [x] OBD connection flow (UI; actual hardware testing in Phase 2)
- [x] Account management page
- [x] Database setup (users, chat, car profiles)
- [x] Claude/GPT-4 API integration
- [x] Basic styling (Tailwind, responsive design)

Deliverable: Working chat app where users can ask questions about car diagnostics.

Phase 2: Post-Car Diagnostics (4-6 weeks)

Goal: Full diagnostic engine + 3D visualization + solutions marketplace

- ☐ OBD Bluetooth integration (real hardware testing)
- ☐ Diagnosis page with 3D car hologram
- ☐ Real-time OBD monitoring
- ☐ Solutions page (mechanic matching)
- ☐ Service reports (AI-generated)
- ☐ Notifications/alerts system
- ☐ Google Maps integration (mechanic locations)
- ☐ WebSocket real-time chat
- ☐ Testing on iOS/Android devices

Deliverable: Users can diagnose car problems, get mechanic recommendations, and book services.

Phase 3: Pre-Car Ownership Journey (3-4 weeks)

Goal: Car research, recommendations, pricing for pre-car buyers

- ☐ Car information library
- ☐ Car recommendation engine
- ☐ Price gauging & market analysis
- ☐ Dealership mapping

- ☐ Journey toggle (user can switch between Pre/Post)
- ☐ Avatar personality adaptation (research mode)
- ☐ Testing pre-car flows

Deliverable: Buyers can research and compare cars before purchase.

Phase 4: Marketplace & Community (4-6 weeks)

Goal: Partner mechanic integration + review system + learning loops

- ☐ Mechanic partner database setup
- ☐ Review/rating system
- ☐ Cost comparison algorithm
- ☐ Mechanic notification system (new service requests)
- ☐ Analytics dashboard
- ☐ Learning feedback loops

Deliverable: Vibrant marketplace with trusted mechanics + continuous improvement.

Phase 5: Custom Model & Optimization (Ongoing)

Goal: Build custom AI model + performance optimization

- ☐ Collect training data from Phase 1-4
- ☐ Fine-tune custom LLM
- ☐ Deploy on own infrastructure
- ☐ Advanced diagnostics (multi-model analysis)
- ☐ Multilingual support (Swahili, French, etc.)

- [] Regional expansion (Kenya, Nigeria, South Africa, etc.)
-

12. SUCCESS METRICS

User Adoption

- Sign-ups per week
- Active users (DAU/MAU)
- Retention rate (day 7, day 30)

Engagement

- Average chats per user per week
- Chat length (avg messages per session)
- Pages visited per session

Functionality

- Diagnosis accuracy (user confirmation of fixes)
- Mechanic match quality (user satisfaction rating)
- Average cost savings (diagnosis vs. blindly replacing parts)

Business

- Mechanic partner growth
 - Average revenue per user (commission on services)
 - Customer acquisition cost (CAC)
 - Lifetime value (LTV)
-

13. GLOSSARY

Term	Definition
OBD	On-Board Diagnostics; reads vehicle fault codes & sensor data via Bluetooth adapter
DTC	Diagnostic Trouble Code (e.g., P0101); identifies specific vehicle problems
TTS	Text-to-Speech; converts text to Avatar voice
STT	Speech-to-Text; converts user voice to text for processing
Phronesis	Greek term for practical wisdom; the app's AI avatar
Pre-Car Journey	User phase before buying a vehicle (research, comparison, pricing)
Post-Car Journey	User phase after owning a vehicle (diagnostics, maintenance, repairs)
Solutions Page	Displays repair recommendations, mechanic options, cost estimates
Diagnosis Report	AI-generated assessment of vehicle problem & repair solutions

14. APPENDICES

A. User Flow Diagram (Pre-Car)

Welcome → Auth → Home/Chat → "Show me cars under \$8k"
→ Car Info Page → "Where can I buy?"
→ Maps Page → [End]

B. User Flow Diagram (Post-Car)

Welcome → Auth → OBD Setup → Home/Chat → "Weird noise"
→ Diagnosis Page → "How much to fix?"
→ Solutions Page → "Where's the shop?"
→ Maps Page → [After service] → Rate Mechanic

C. Avatar State Machine

State	Trigger	Next State	Action
Welcome	Page Load	Listening	Ask "Do you own a car?"
Listening	User Response	Routing	Infer journey, personalize greeting
Routing	Intent Detected	Page Navigation	Navigate to appropriate page
Diagnostics	"Weird noise"	Analysis	Analyze OBD + user input
Analysis	Analysis Complete	Solutions	Show repair options

State	Trigger	Next State	Action
Solutions	User Select Shop	Navigation	Route to Maps

15. NEXT STEPS

1. **Review & Approval:** Stakeholder sign-off on this plan
2. **Technical Design:** Create detailed API contracts, database schemas, UI wireframes
3. **Environment Setup:** Supabase project, API keys, Git repo structure
4. **Sprint Planning:** Break Phase 1 into 2-week sprints
5. **Recruitment:** Hire/assign developers, designers, QA testers
6. **Kickoff:** Start Phase 1 development

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